DEMAND AND SUPPLY

THE CONCEPT OF MARKET

The word 'market' generally means a place or area where goods and services are bought and sold, e.g., Chandani Chowk, Karol Bagh, Connaught Place, Delhi Stock Exchange, Bombay Stock Exchange, Sabzi Mandi. In economics, however, the word 'market' is used rather in an abstract sense. The market refers to a system by which sellers and buyers of a commodity interact to settle its price and the quantity to be bought and sold. According to Samuelson and Nordhaus (1995), 'A market is a mechanism by which buyers and sellers interact to determine the price and quantity of a good or service. The buyers and sellers may be individuals, firms, factories, dealers and agents. Buyers constitute the demand side and sellers constitute the supply side of the market'.

Following are some conceptual aspects of the market system:

1. A market need not be situated in a particular place or locality. The geographical area of a market depends on the area over which the buyers and sellers are spread. It may be as small as a fish market in a corner of the city or as large as the entire world, e.g., the global markets for arms, cars, electronic goods, aeroplanes, computers, petroleum products, medicines.
2. Buyers and sellers need not come in personal contact with each other. The transactions can be carried out through postal services, telephone, fax, agents, or e-mail, etc. People do buy many goods and services directly from the producers by telephone calls or e-mails without having seen them ever.
3. The word 'market' may refer to a commodity or service (e.g., fruit market, car market, share market, money market, paper market, labour market) or to a geographical area (Bombay market, Indian market, Asian market, etc.).
4. The economists distinguish between the markets also on the basis of (a) nature of goods and services, e.g., factor market and commodity market or input market and output market and (b) number of firms and degree of competition, e.g., competitive market (large number of firms with homogenous products), monopolistic market (many firms with differentiated products), oligopolistic market.

THE DEMAND SIDE OF THE MARKET

Meaning of Demand

Conceptually, demand can be defined as the desire to buy a good for which the demander has ability and willingness to pay. In simple words, demand is a desire for a good, backed by ability and willingness to pay. A desire without ability to pay is merely a wish. A desire with ability to pay but without willingness to pay is only a potential demand.

A desire accompanied by ability and willingness to pay makes a real or effective demand.

- ***Individual and Market Demand.*** For the purpose of demand analysis, a distinction is often made between the *individual demand* and the *market demand*—individual demand for analysing consumer behaviour and market demand for analysing market behaviour.
- **Individual Demand**. Can be defined as the quantity of a commodity that a person is willing to buy at a given price over a specified period of time, say per day, per week, per month, etc.
- **Market Demand**. Refers to the total quantity that all the users of a commodity are willing to buy at a given price over a specific period of time. In fact, market demand is the sum of individual demands for a product. Individual and market demand curves are discussed ahead in detail. Let us now discuss the law of demand. Since we are concerned in this chapter with market demand, The law of demand will be discussed in the context of market demand.

The Law of Demand

The law of demand states the relationship between the quantity demanded and the price of a commodity. In general, quantity demanded of a commodity depends on many other factors also, viz., consumer's income, price of the related goods (substitutes and complements), consumer's taste and preferences, advertisement, etc. However, price of a good is the most important and the only determinant of its demand in the *short run* because other factors remain constant. Therefore, the law of demand is linked to the price of the product.

The law of demand can be stated as 'all other things remaining constant, the quantity demanded of a commodity increases when its price decreases and decreases when its price increases'. This law implies that demand and price are inversely related. Marshall states the law of demand as 'the amount demanded increases with a fall in price and diminishes with a rise in price'. This law holds under *ceteris paribus* assumption, i.e., *all other things remain unchanged*. The law of demand can be illustrated through a *demand schedule* and a *demand curve*.

The Demand Schedule

A demand schedule is a tabular presentation of quantity demanded of a commodity at different prices per unit of time. A hypothetical market demand schedule is given in Table 3.1. This table presents price of shirts (Ps) and the corresponding number of shirts demanded (Qs) per month. Table 3.1 illustrates the law of demand. As data given in the table shows, the demand for shirts (Qs) increases as its price (Ps) decreases. For instance, at price Rs 800 per shirt, only 10,000 shirts are demanded per month. When price decreases to Rs 400, the demand for shirts increases to 30,000 and when price falls further to Rs 100, demand rises to 80,000. Similarly, one can read the table in reverse order and arrive at the conclusion that as price of shirt increases, its demand decreases. This relationship between the price and the quantity demanded gives the law of demand.

Table 3.1 *Demand Schedule for Shirts*

P_s (price in Rs)	Q_s (shirts in '000)
800	8
600	15
400	30
300	40
200	55
100	80

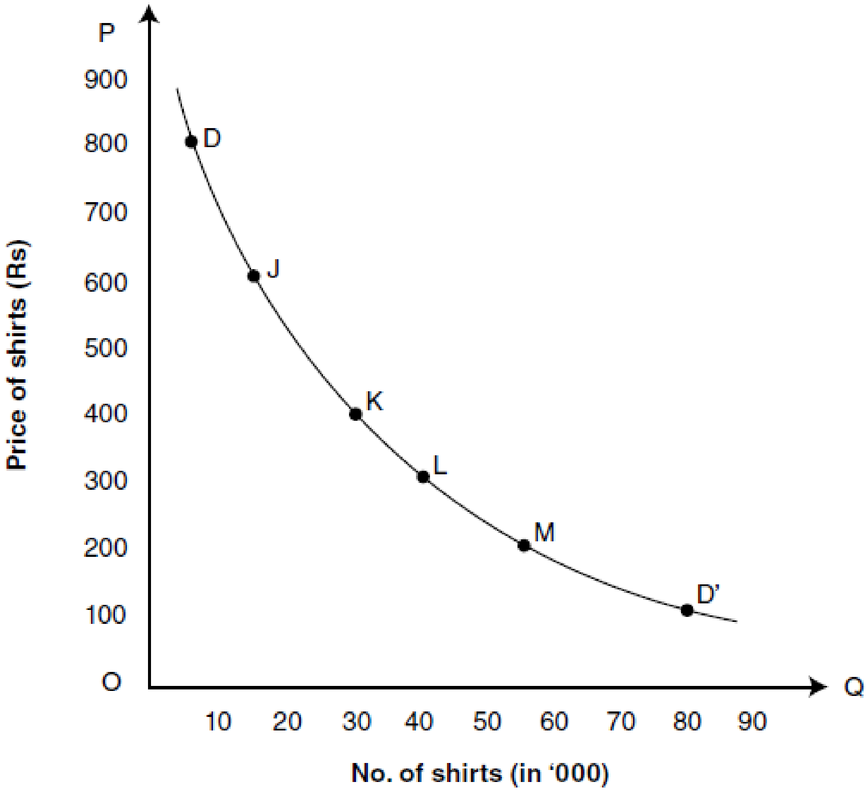


Figure 3.1 *The Demand Curve*

The Demand Curve

A *demand curve* is a graphical presentation of the demand schedule. For example, when the data given in the demand schedule (Table 3.1) are presented graphically as shown in Figure 3.1, the resulting curve *DD′* is the *demand curve*. The curve *DD′* in Figure 3.1 depicts the law of demand. It slopes downward to the right. That is, it has a negative slope. The negative slope of the demand curve *DD′* shows the *inverse relationship* between the price of shirt and its quantity demanded. The inverse relationship means that demand increases with the decrease in price and decreases with the rise in price. As can be seen in Figure 3.1, downward movement on the demand curve *DD′* from point *D* towards *D′* shows fall in price and rise in demand. Similarly, an upward movement from point *D′* towards *D* reads rise in price and fall in demand.

The law of demand is based on an empirical fact. For example, when prices of cell phones and personal computers (PCs), especially of the latter, were astronomically high, only few rich persons and big firms could afford them. Now with the revolution in computer and cell phone technology and the consequent fall in their prices, demand for these goods has shot up in India though other factors too contributed to rise in demand for these goods.

The Factors Behind the Law of Demand

According to the law of demand, when a price of a product increases, its demand decreases and *vice versa*, all other things remain the same. A question arises here: what are the factors behind the law of demand or why does demand decrease when price rises or other way round? The factors *behind the law of demand* are the following.

1. ***Income Effect.*** When the price of a commodity falls, real income of its consumers increases in terms of this commodity. In other words, their purchasing power increases since they are required to pay less for the same quantity. According to another economic law, increase in real income (or purchasing power) increases demand for the goods and services in general and for the goods with reduced price in particular. The increase in demand on account of increase in real income is called *income effect*. It should, however, be noted that the *income effect* is negative in case of *inferior goods*. In case, price of an inferior good accounting for a considerable proportion of the total consumer expenditure falls substantially, consumers' real income increases. Consequently, they substitute superior goods for inferior ones. Therefore, *income effect* on the demand for the inferior good becomes *negative*.
2. ***Substitution Effect.*** When price of a commodity falls, it becomes cheaper compared to its substitutes, prices of substitutes remaining constant. In other words, when price of a commodity falls, price of its substitutes remaining the same, its substitutes become relatively costlier. Consequently, rational consumers tend to substitute cheaper goods for costlier ones within the range of normal goods—goods whose demand increases with the increase in consumer's income—other things remaining the same. Therefore, demand for the relatively cheaper goods increases. The increase in demand on account of this factor is known as *substitution effect*.
3. ***Diminishing Marginal Utility.*** Marginal utility is the utility derived from the marginal unit consumed of a commodity. Diminishing marginal utility is also responsible for the increase in demand for a commodity when its price falls. When a person buys a commodity, he exchanges his money income with the commodity in order to maximize his satisfaction. He continues to buy goods and services so long as marginal utility of his money (MUm) is less than the marginal utility of the commodity (MUc). Given the price (Pc) of a commodity, the consumer adjusts his purchases so that MUc = MUm. This proposition holds under both constant MUm and diminishing MUm.

If MUm is assumed to be constant, then MUm = Pc. Under this condition, utility-maximizing consumer makes his purchases in such quantities that
MU= P= MU m c c

When price falls, (MUm = Pc) < MUc. The only way to regain his equilibrium is to reduce MUc. So the consumer purchases more of the commodity. When the stock of a commodity increases, MUc decreases. As a result, demand for a commodity increases when its price decreases.

This conclusion holds also under diminishing MUm. When price of a commodity falls and consumer buys only as many units as before the fall in price, he saves some money on this commodity. As a result, his stock of money increases and his MUm decreases, whereas MUc remains unchanged because his stock of commodity remains unchanged. Since MUm is less than MUc the utility-maximizing consumer exchanges money with commodity to equate MUm with MUc, with a view to maximizing his satisfaction. Consequently, demand for a commodity increases when its price falls.

Exceptions to the Law of Demand

The law of demand is one of the fundamental laws of economics. The law of demand, however, does not apply to the following cases.

1. ***Expectations Regarding Future Prices.*** When consumers expect a continuous increase in the price of a durable commodity, they buy more of it, despite the increase in its price, to avoid the pinch of still higher price in future. Similarly, when consumers anticipate a considerable decrease in the price in future, they postpone their purchases and wait for the price to fall further, rather than buy the commodity when its price initially falls. Such decisions of the consumers are contrary to the law of demand.
2. ***Prestigious Goods.*** The law does not apply to the commodities which are used as a 'status symbol', which enhance social prestige or display wealth and richness, e.g., gold,2 precious stones, rare paintings and antiques. Rich people buy such goods mainly because their prices are high.
3. ***Giffen Goods.*** A classic exception to the law of demand is the case of Giffen goods named after a British economist, Sir Robert Giffen (1837–1910). A Giffen good does not mean any specific commodity. It may be any inferior but essential commodity much cheaper than its substitutes, consumed mostly by the poor households and

claiming a large part of their income. If the price of such goods increases (price of its substitute remaining constant), its demand increases instead of decreasing. For instance, let us suppose that the *monthly minimum* consumption of food grains by a poor household is 30 kg including 20 kg of *bajra* (an inferior good) and 10 kg of wheat (a superior good). Suppose also that *bajra* sells at Rs 5/kg and wheat at Rs 10/kg. At these prices, the household spends Rs 200 per month on food grains. That is the maximum it can afford. Now, if price of *bajra* increases to Rs 6 per kg, the household will be forced to reduce its consumption of wheat by 5 kg3 and increase that of *bajra* by the same quantity in order to meet its minimum monthly consumption requirement within Rs 200 per month. Obviously, household's demand for *bajra* increases from 20 to 25 kg per month despite increase in its price and that of wheat falls to 5 kg

The Market Demand

As already mentioned in Section 'Meaning of Demand', *market demand* for a commodity is the sum of all individual demands for the commodity at a given price, per unit of time. In this section, we explain the concept of market demand in detail and illustrate the derivation of the market demand curve.

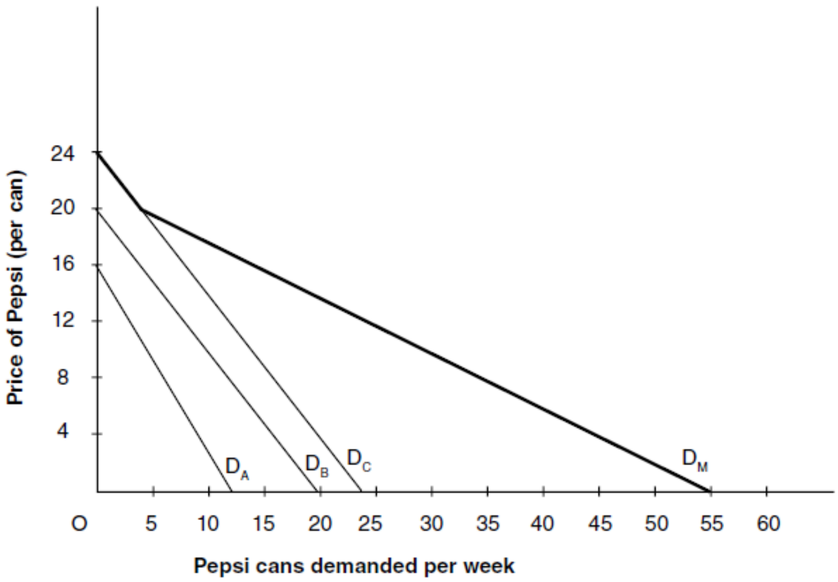
Suppose, there are only three consumers (*A, B* and *C*) of Pepsi and their weekly individual demand for Pepsi at its different prices is given in Table 3.2. The last column of the table shows the market demand, i.e., the sum of individual weekly demands for Pepsi cans.

Table 3.2 *Weekly Individual and Market Demand for Pepsi Cans* Price

Price (Rs)	No. of Pepsi Cans Demanded By			Market Demand $= A + B + C$
	A	*B*	*C*	
16	0	4	8	12
12	3	8	12	23
8	5	12	16	33
4	8	16	20	44
0	11	20	24	55

Graphical Derivation

The market demand curve can be drawn straightaway by plotting the data in the last column of Table 3.2. Alternatively, market demand curve can be derived graphically by horizontal summation of the individual demand curves at each price of the Pepsi cans. Graphical derivation of the market demand curve is illustrated in Figure 3.2. The individual demand curves of buyers *A, B* and *C* are shown by the demand curves *DA*, *DB* and *DC*, respectively. Horizontal summation of these demand curves produces weekly *market demand curve* for Pepsi cans as shown by the curve *D*M. Thus, graphically, a *market demand curve is horizontal summation of individual demand curves at different prices*. It is *important* to note here that there is a significant difference between the individual demand curves and the market demand curves. The individual demand curves may not slope downward in case of many seasonal and occasional consumer goods, e.g., a book by an author, an umbrella, a cinema ticket for a show, or a passenger ticket. But market demand for all such goods slopes downward following the decrease in their prices because the fall in price causes increase in the number of consumers. In other words, even if individual demands are in the form of vertical lines, market demand curve slopes downward to the right.



Demand Function

In mathematical language, a function is a symbolic statement of relationship between two or more Interrelated variables – generally one dependent and one or more independent variables. Demand function states the relationship between demand for a product (the dependent variable) and its determinants (the independent variables). Let us consider the most common form of a demand function, i.e., the short-run demand function, which consists of quantity demanded of a product (*D*) – the dependent variable – and price of the product (*P*) – the independent variables. Assume that the quantity demanded of a commodity *X* (*Dx*) depends only on its price (*Px*), other factors remaining constant. The demand function will then read as 'demand for a commodity (*Dx*) depends on its price (*Px*)'. The same statement may be written in its functional form as

$Dx = f\,(Px)$ (3.1)

where *Dx* is demand for commodity *X* (the dependent variable) and *Px* is price of *X* (the independent
variable)

The function (3.1), however, states only that there is a relationship between quantity demanded of commodity *X* and its price. More specifically, it states that the quantity demanded of *X* depends on its price. It does not give the quantitative relationship between *Dx* and *P*

Shift in Demand Curve

A short-run demand curve is drawn on the basis of a demand schedule or a short-run demand function. So long as price–demand relationship is fixed, the demand curve remains fixed at its position, as shown in Figure 3.7. However, given the price–demand relationship, if there is a change in other determinants of demand, e.g., in consumer's income and in the price of substitute or complementary goods, the demand curve may shift upward or downward depending on the direction of the change in other determinants. Let us suppose that the demand curve, *D2*, in Figure 3.9 is the original demand curve for commodity *X*.

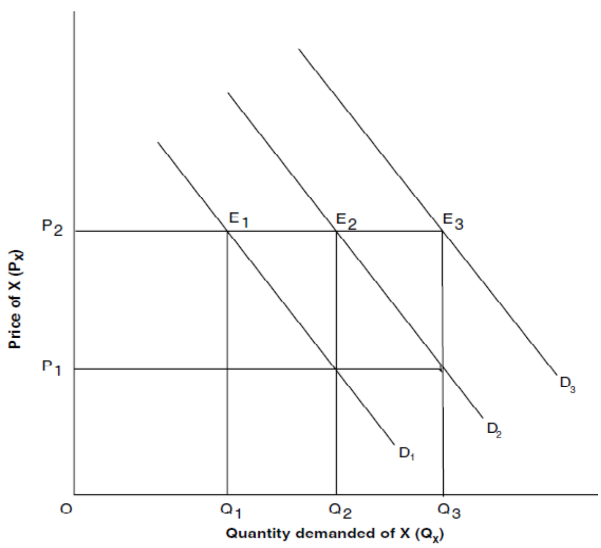


Figure 3.9 *Shift in Demand Curve*

As shown in the figure, at price *OP*2, demand equals *OQ*2 units of *X*, other factors remaining constant. But if any of the other factors (e.g., consumers' income or price of the substitutes) changes, it will change consumer's ability and willingness to buy commodity *X*. For example, if consumer's disposable income decreases due to, say, increase in income tax, he may be able to buy only *OQ*1 units of *X* instead of *OQ*2 at price *OP*2. As a result, demand curve *D*2 shifts downward *D*1. This is true for the whole range of price of *X*, that is, consumers would be able to buy less at all other prices. This will cause a *downward shift* in demand curve from *D*2 to *D*1.

Similarly, increase in disposable income of the consumer, say, due to reduction in taxes, may cause an *upward shift* from *D*2 to *D*3. For example, suppose consumer is initially at equilibrium at point *E*2 on demand curve *D*2.

Let consumer's income increase, all other factors remaining constant, so that he can buy more of commodity
X. As a result, the consumer shifts to point *E*3 on demand curve *D*3 and can buy *OQ*3 units of commodity
X. This condition applies to all the levels of prices. So the demand curve shifts from the position of *D*2 to *D*3.
Such changes in the location of demand curves are known as shifts in demand curve.

Reasons for Shift in Demand Curve

Shifts in a demand curve may take place owing to the change in one or more of the determinants of demand. Consider, for example, the decrease in demand for commodity *X* by *Q*1*Q*2 in Figure 3.8. This fall in demand may have been caused by any or many of the following reasons:

1. Fall in the consumer's income so that consumer can buy only *OQ*1 of *X* at price *OP*2—it is called *income effect*;
2. Fall in the price of substitute of commodity *X* so that the consumers find it gainful to substitute *Q*1*Q*2 of *X* for its substitute—it is *substitution effect*;
3. Advertisement made by the producer of the substitute changes consumer's taste or preference against commodity *X* so much that they replace *Q*1*Q*2 of it with its substitute—again a *advertisement effect*;

4. Increase in the price of complements of *X* so that consumers can afford only *OQ*1 of *X*. This is *price effect* of the complementary good; and
5. Other reasons such as commodity *X* is going out of fashion, its quality has deteriorated, consumers' technology has so changed that only *OQ*1 of *X* can be used, and change in season if commodity *X* has only seasonal use.

THE SUPPLY SIDE OF THE MARKET

In a market economy, while buyers of a product constitute the demand side of the market, sellers of that product make the supply side of the market. In this section, we discuss the supply side of the market.

Market Supply

Supply Means the Quantity of a Commodity Which Its Producers or Sellers Offer for Sell at a Given Price, Per Unit of Time Market supply, like market demand, is the sum of supplies of a commodity made by all individual firms or suppliers.

The Law of Supply

In general sense of the term, the supply of a commodity depends on its price. In other words, supply of a product is the function of its price. The law of supply is expressed generally in terms of price–quantity relationship. The *law of supply* can be stated as follows: ***The supply of a product increases with the increase in its price and decreases with decrease in its price, other things remaining constant***. It implies that the supply of a commodity and its price are positively related. This relationship holds under the assumption that "other things remain the same". "Other things" include technology, price of related goods (substitute and complements), consumers' taste and preferences, and weather and climatic conditions in case of agricultural products.

The Supply Schedule and Supply Curve

The law of supply can be depicted by a supply schedule and a supply curve. A supply schedule is a table showing different prices of a commodity and the corresponding quantity that suppliers are willing to offer for sale. Table 3.3 presents a hypothetical supply schedule of shirts, i.e., number of shirts supplied per month at different prices.

The supply curve is a graphical presentation of the supply schedule. The supply curve *SS*′ given in Figure 3.10 has been drawn by plotting the price–supply data given in Table 3.3. The points *S, P, Q, R, T* and *S*′ show the price–quantity combinations on the supply curve *SS*′. The supply curve, *SS*′, depicts *the law of supply*. The upward slope of the supply curve indicates the rise in the supply of shirts with the rise in its price and *vice versa*. That is, the supply of shirts increases with the rise in its price and *vice versa*. For example, at price Rs 200, only 35,000 shirts are supplied per month. When price rises to Rs 400, supply increases to 60,000 shirts.

Table 3.3 *Supply Schedule of Shirts*

Price (in Rs)	Supply (shirts in '000)
100	10
200	35
300	50
400	60
600	75
800	80

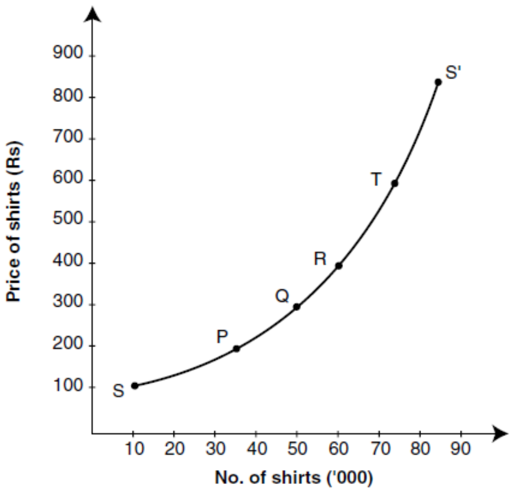


Figure 3.10 *Supply Curve of Shirts*

As shown in Figure 3.10, *a supply curve has a positive slope*. The positive slope of the supply curve is caused by seller's desire to make larger profit and, more importantly, by the rise in cost of production. In fact, when price of a commodity increases, its suppliers tend to supply more and more. To supply more and more, they need to produce more and more. When they increase production, cost of production increases due to the law of diminishing returns. In fact, supply curve is derived from the marginal cost curve. (The derivation of supply curve on the basis of the marginal cost curve is discussed in detail and illustrated ahead in Chapter 15.)

Shift in the Supply Curve

We have shown above that a change in the price of a commodity causes a change in its quantity supplied along a given supply curve. Although price of a commodity is the most important determinant of its supply,

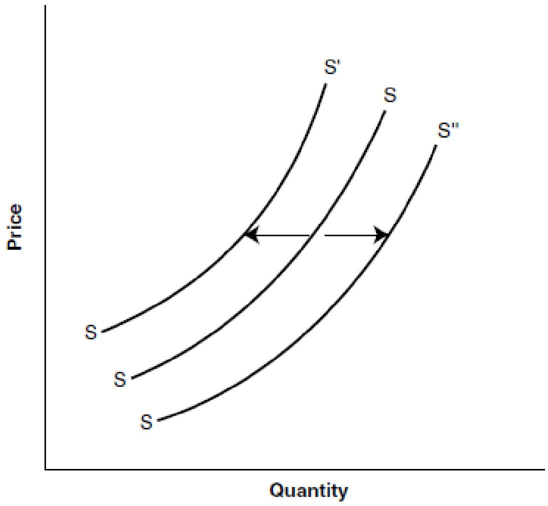


Figure 3.11 *Shift in the Supply Curve*

it is not the only determinant. Many other factors influence the supply of a commodity. Given the supply curve of a commodity, when there is a change in its other determinants, the supply curve shifts rightward or leftward depending on the effect of such changes. Let us now explain how other determinants of supply cause shift in the supply curve.

1. ***Change in Input Prices.*** Input prices include the price of labour, raw materials, overheads, etc. When input prices decrease, the use of inputs increases. As a result, product supply increases and the supply curve *SS* shifts to the right to *SS*″, as shown in Figure 3.11. Similarly, when input prices increase, product supply curve shifts leftward from *SS* to *SS*′.
2. ***Technological Progress.*** Technological progress reduces cost of production or increases labour productively or do both. Technological progress that reduces cost of production or increases efficiency causes increase in product supply. For instance, introduction of high-yielding variety of paddy and new techniques of cultivation increased per-acre yield of rice in India in the 1970s. Such changes make the supply curve shift to the right.
3. ***Product Diversification and Cost Reduction.*** In production of many commodities, it is possible to produce some other goods which require a similar technology. For example, a refrigerator company can also produce ACs; Tatas famous for truck production can also produce Nano and other types of cars; Maruti Udyog can produce

trucks and so on. Product diversification may cause reduction in the production cost of the main product. This may lead to the rise in the supply of the main product due to capacity utilization for profit maximization.

4. ***Nature and Size of the Industry.*** The supply of a commodity depends also on whether an industry is monopolized or competitive. Under monopoly, supply of a product is shorter than it is in a competitive market. When a monopolized industry is made competitive, the total supply increases. Besides, if size of an industry increases due to new firms joining the industry, the total supply increases and supply curve shifts rightward.
5. ***Government Policy.*** When government imposes restrictions on production, e.g., import quota on inputs, rationing of or quota imposed on input supply, etc., production tends to fall. Such restrictions make supply curve to shift leftward.
6. ***Non-Economic Factors.*** The factors like labour strikes and lock-outs, war, droughts, floods, communal riots, epidemics, etc. also affect adversely the supply of commodities making supply curve shift leftward.

Supply Function

A *supply function* is a mathematical statement which states the relationship between the quantity supplied of a commodity and its determinants. The short-run market supply function is based on the law of supply. The law of supply states only the nature of relationship between the price and the quantity supplied, i.e., supply increases with the increase in price. A supply function that quantifies this relationship is written as

$Qx = dPx$ (3.9a)

where Qx denotes the quantity supplied of commodity X, Px denotes its price and d gives the measure of relationship between Qx and Px.

Once the relationship between Qx and Px is measured in numerical terms, i.e., the numerical value of 'd' is known, then the supply function can be expressed numerically. For example, suppose $d = 10$, then the supply function can be expressed as

$Qx = 10Px$ (3.9b)

Given the supply function (3.9b), a supply schedule can be obtained by substituting numerical values for Px. For example, if $Px = 2$, $Qx = 20$ and if $Px = 5$, $Qx = 50$. By plotting the supply schedule, a supply curve can be obtained. (For procedure, refer to the section on demand function).

THE MARKET EQUILIBRIUM: THE EQUILIBRIUM OF DEMAND AND SUPPLY

Determination of Price in a Free Market

In sections 'The Demand Side of the Market' and 'The Supply Side of the Market', we have explained the laws governing the market forces demand and supply — and how demand and supply behave in response to the change in price and other determinants. In this section, we explain how demand and supply strike a balance, how market attains *equilibrium*, and how *equilibrium price* is determined in a *free market*. A free market is one in which market forces of demand and supply are free to take their own course and there is no outside control on price, demand and supply.

The Concept of Market Equilibrium

In physical sense, the term equilibrium means the 'state of rest'. In general sense, it means balance in opposite forces. In the context of market analysis, *equilibrium refers to a state of market in which quantity demanded of a commodity equals the quantity supplied of the commodity*. The equality of demand and supply produces an *equilibrium price*. The *equilibrium price* is the price at which quantity demanded of a commodity equals its quantity supplied. That is, at equilibrium price, demand and supply are in balance. Equilibrium price is also called *market-clearing price*. Market is cleared in the sense that there is no unsold stock and no unsupplied demand.

Table 3.4 *Monthly Demand and Supply Schedules for Shirts*

Price per Shirt (Rs)	Demand ('000 shirts)	Supply ('000 shirts)	Market Position	Effect on Price
100	80	10	Demand exceeds supply	Rise
200	55	28	Demand exceeds supply	Rise
300	40	40	Equilibrium	Stable
400	28	50	Supply exceeds demand	Fall
500	20	55	Supply exceeds demand	Fall
600	15	60	Supply exceeds demand	Fall

Determination of Market Price

The equilibrium price of a commodity in a free market is determined by the market forces of demand and supply. In order to analyse how equilibrium price is determined, we need to integrate the demand and supply curves. For this purpose, let us use the example of shirts. Let us suppose that the market demand and supply schedules for shirts are given as shown in Table 3.4.

As the table shows, there is only one price of shirts (Rs 300) at which the market is in equilibrium, i.e., the quantity demanded equals the quantity supplied at 40,000 shirts per month. At all other prices, the shirt market is in *disequilibrium*. That is, there is imbalance between supply and demand. When market is in the state of disequilibrium, either demand exceeds supply or supply exceeds demand. As the table shows, at all prices below Rs 300 per shirt, demand exceeds supply showing *shortage* of shirts in the market. Likewise, at all prices above Rs 300, supply exceeds demand showing *excess supply*.

In a Free Market, Disequilibrium Itself Creates the Condition for Equilibrium

When there is excess supply, it means unsold stock. The unsold stock causes a loss to the firms. This forces firms to cut down their supply and price. Thus, excess supply itself forces downward adjustments in the price and the quantity supplied. The process of downward adjustments continues till supply equals demand.

Similarly, when there is excess demand, it forces upward adjustments in the price and quantity demanded. When there is excess demand, firms take the advantage of the market situation and increase supply. When they increase production, cost of production goes up. But consumers, given their demand curve, are willing to pay a higher price. This process continues until demand equals supply. In our example, the process of downward and upward adjustments in price and quantity continues till the price reaches Rs 300 and quantities supplied and demanded per month balance at 40,000 shirts. This process is automatic. Let us now look into the process of price and quantity adjustments called *market mechanism*.

Market Mechanism: How Market Brings About Balance

Market mechanism is a process of interaction between the market forces of demand and supply to determine equilibrium price. To understand how it works, let us explain the process through our own
example. Suppose price of the shirts be initially set at Rs 100. As shown in Table 3.4, at this price, the quantity demanded of shirts is 80,000 and the quantity supplied is 10,000 shirts. Thus, at price Rs 100 per shirt, demand exceeds the quantity supplied by 70,000 shirts. The shortage will force buyers to bid higher price to buy the desired number of shirts. This gives

sellers an opportunity to raise the price. Increase in price enhances the profit margin. This induces firms to produce and sell more in order to maximize their profits. This trend continues till price rises to Rs 300. As Table 3.4 shows, at price Rs 300, the buyers are willing to buy 40,000 shirts. This is exactly the number of shirts that sellers would like to sell at this price.

At this price, there is neither shortage nor excess supply of shirts in the market. Therefore, Rs 300 is the equilibrium price. The market is, therefore, in equilibrium. Similarly, at all prices above Rs 300, supply exceeds demand showing excess supply of shirts in the market. The excess supply forces the competing sellers to cut down the price. Some firms find low price unprofitable and go out of market and some cut down their production. Therefore, supply of shirts goes down. On the other hand, fall in price invites more customers. This process continues until price of shirts falls to Rs 300. At this price, demand and supply are in balance and market is in equilibrium. Therefore, the price at Rs 300 per shirt is equilibrium price.

Graphical Illustration of Price Determination

The determination of equilibrium price is illustrated graphically in Figure 3.12. The demand curve *DD'* and the supply curve *SS'* have been drawn by plotting the demand and supply schedules, respectively, (given in Table 3.4) on the price and quantity axes.

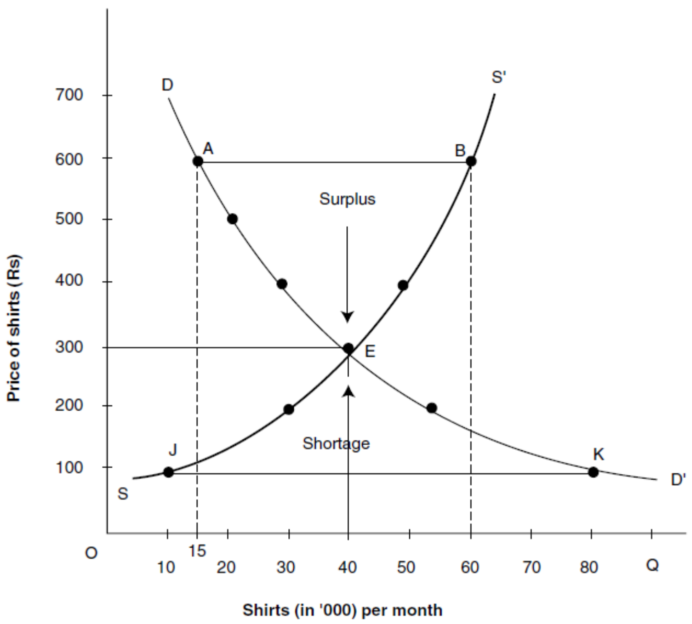


Figure 3.12 *Equilibrium of Demand and Supply: Price Determination*

As Figure 3.12 shows, demand and supply curves intersect at point *E* determining the *equilibrium price* at Rs 300. At this price, the quantity demanded (40,000 shirts) per month equals the quantity supplied. Thus, the equilibrium price is Rs 300 and the equilibrium quantity is 40,000 shirts. The equilibrium condition is not fulfilled at any other point on the demand and supply curves. Therefore, if price is set at any price other than Rs 300, there would be either excess supply or excess demand for shirts in the market.

Let us now see how market works to bring about balance in demand for and supply of shirts. Let the price be initially set at Rs 600. At this price, suppliers bring in a supply of 60,000 shirts, whereas buyers are willing to buy only 15,000 shirts. The supply, obviously, far exceeds the demand. As Figure 3.12 shows, the excess supply equals, *AB* = 60,000 − 15,000 = 45,000 shirts. The suppliers would, therefore, lower down the price gradually in order to get rid of the unsold stock and cut down the supply simultaneously. Besides, when price falls, demand for shirts increases too. In this process, the supply–demand gap is reduced. This process continues until price reaches Rs 300 at point *E*, the point of equilibrium where demand and supply equal at 40,000 shirts. At this price, the market is in equilibrium and there is no inherent force at work which can disturb the market equilibrium.

Likewise, if price is initially set at Rs 100, the buyers would be willing to buy 80,000 shirts, whereas suppliers would be willing to supply only 10,000 shirts. Thus, there would be a shortage of 70,000 shirts as shown by the distance *JK* in Figure 3.12. The shortage will force the buyers to bid a higher price. This will lead to increase in price which will encourage the suppliers to increase their supply. This process of adjustment will continue as long as demand exceeds supply. When price rises to Rs 300, the market reaches its equilibrium.

SHIFT IN DEMAND AND SUPPLY CURVES AND MARKET EQUILIBRIUM

Shift in Demand Curve

Whenever there is a shift in the demand and/or supply curve, there is also a shift in the equilibrium point. The effect of shift in the demand curve on the equilibrium is shown in Figure 3.14(a). Suppose that the initial demand curve is given by the curve *DD*1 and supply curve by *SS*. The demand and supply curves intersect at point *P*. The equilibrium price is determined at *PQ* and equilibrium quantity at *OQ*. Let the demand curve now shifts from its position *DD*1 to *DD*2, supply curve remaining the same.

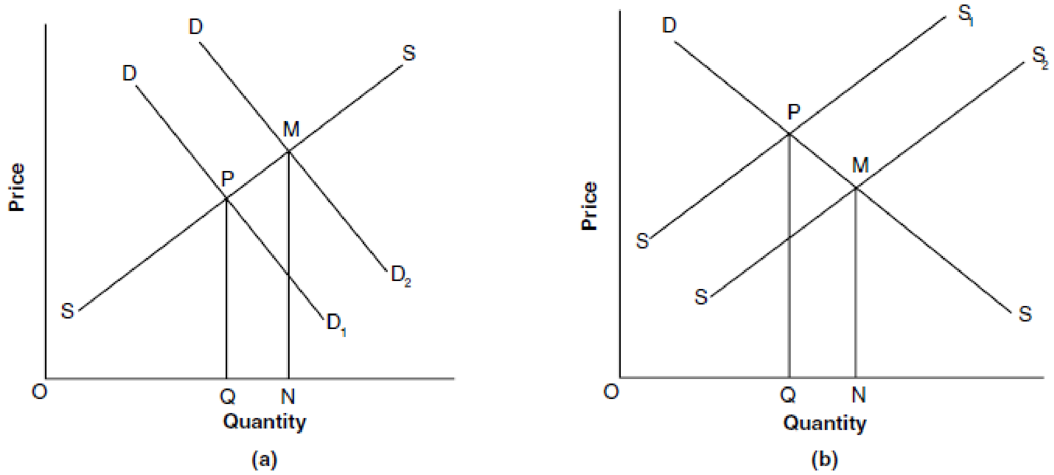


Figure 3.14 **(a)** *Shift in Demand Curve and Equilibrium and* **(b)** *Shift in Supply Curve and Equilibrium*

The demand curve *DD*2 intersects the supply curve *SS*′ at point *M*. Thus, shift in the demand curve causes a shift in the equilibrium from point *P* to point *M*. At the new equilibrium, the quantity demanded and supplied increases from *OQ* to *ON* and the price increases from *PQ* to *MN*. Note that, the supply curve remaining the same, a rightward shift in the demand curve results in a higher equilibrium price and a higher equilibrium quantity. A downward shift in the demand curve from *DD*2 to *DD*1 produces a reverse result—fall in both the equilibrium price and the quantity demanded and supplied.

Shift in Supply Curve

Figure 3.14(b) shows the effect of shift in the supply curve on the equilibrium, demand curve remaining the same. Suppose that the demand curve is given as *DD* and initial supply curve as *SS*1. The curves *DD* and *SS*1 intersect at point *P*, determining equilibrium price at *PQ* and equilibrium demand and supply at *OQ*. Let the supply curve now shifts from its position *SS*1 to *SS*2, demand curve remaining unchanged. The new supply curve *SS*2 intersects the demand curve at point *M*. Thus, a new equilibrium takes place at point *M* where equilibrium price is *MN* and equilibrium quantity is *ON*. Note that a rightward shift in the supply curve, demand curve remaining the same, causes equilibrium price to fall and output to increase.

Parallel Shift in Demand and Supply Curves

We have seen above that a rightward shift in the demand curve causes a *rise* in market price (Figure 3.14(b)), and a rightward shift in the supply curve (Figure 3.14(b)) causes a *fall* in the market price. Let us now look at the effect of simultaneous and parallel shifts in demand and supply curves on the equilibrium price and output. The effect of a simultaneous and parallel rightward shift in demand and supply curves on the equilibrium price and output depends on how big or small is the relative shift in demand and supply curves. The *simultaneous and parallel* shifts in demand and supply curves in different measures and its effect on equilibrium price and output are illustrated in panels (a) and (b) of Figure 3.15.

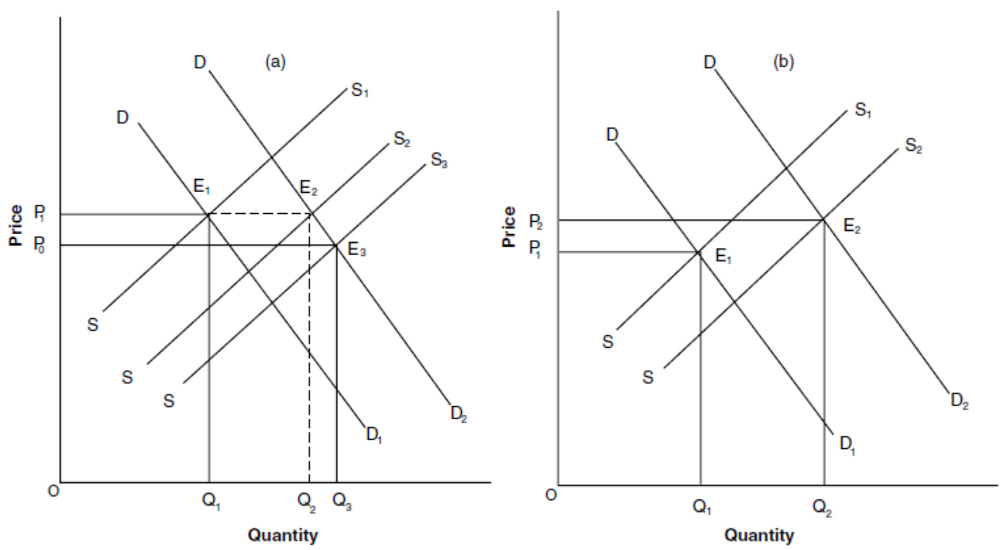


Figure 3.15 *Parallel Shift in Demand and Supply Curves and Its Effect on the Equilibrium Price and Output*

Figure 3.15(a) shows (i) a parallel and equal shift in both demand and supply curves and (ii) a larger shift in supply curve than the demand curve and their effect of the market price and the output.

To explain it further, let us suppose that the initial demand and supply curves are given as *DD*1 and *SS*1, respectively. These demand supply curves intersect at point *E*1 determining the equilibrium price at *P*1 and the equilibrium output at *Q*1. Now let us suppose that there is a parallel and equal shift in both demand and supply curves—demand curve from *DD*1 to *DD*2 and supply from *SS*1 to *SS*2. The demand curve *DD*2 and the supply *SS*2 intersect at point *E*2 determining a new and a greater equilibrium output (*Q*2), while price remains constant at the previous level (*P*1). This means that *when there is a parallel and equal shift in the demand and supply curves, price remains the same but output increases*. Let us now see what happens when the shift in the supply curve is greater than that in the demand curve. Let us suppose that demand curve shifts from *DD*1 to *DD*2 and supply from *SS*1 to *SS*3. Note that the shift in the supply curve is obviously greater than that in the demand curve. As a result, market equilibrium shifts from point *E*1 to *E*3 determining equilibrium price at *E*0 and equilibrium output at *Q*3. Note that a greater shift in the supply curve than the demand curve results in a lower price and a much greater equilibrium output.

Figure 3.15(b) shows the effect of a larger shift in the demand curve than that in the supply curve. Suppose initial demand curve is given as *DD*1 and the supply curve as *SS*1. As the figure shows, the original demand and supply curves intersect at point *E*1 determining the equilibrium price at *P*1 and the equilibrium output at *Q*1. Now let the demand curve shifts from *DD*1 to *DD*2 and the supply curve shifts from *SS*1 to *SS*2. Note that the shift in the demand curve is much larger than the shift in the supply curve. The new demand and supply curves intersect at point *E*2 determining the equilibrium price at *P*2 and the equilibrium output at *Q*2. Note that if the upward shift in the demand curve is greater than that in the supply curve, both equilibrium price and output increase. These are theoretical propositions often so observed in real life.